

CURRICULUM VITAE

Lukas Niebel

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Academic appointments

- from 08/2026 **Postdoctoral researcher**, *ETH Zürich*, Mentor: Prof. Dr Mikaela Iacobelli
04/2023–06/2026 **Postdoctoral researcher**, *University of Münster*, Mentor: Prof. Dr Christian Seis
04/2019–03/2023 **Research associate**, *Ulm University*

Education

- 04/2019–06/2023 **PhD in Mathematics**, *Ulm University*
Thesis: Analytic aspects of kinetic partial differential equations
Advisor: Prof. Dr Rico Zacher
Referees: Prof. Dr Cyril Imbert, Prof. Dr Hongjie Dong
Grade: Summa cum laude; awarded by the Ulmer Universitätsgesellschaft
- 03/2018–03/2019 **MSc in Mathematics**, *Ulm University*
Thesis: Kolmogorov equations – Well-posedness, regularity, asymptotics and Harnack inequalities
Grade: 1.0 with distinction; received the award for the best graduating student
- 10/2014–02/2018 **BSc in Mathematics**, *Ulm University*
Thesis: Long-time behaviour of Markov chains by discrete functional inequalities and entropic Ricci curvature
Grade: 1.1 with distinction; received the award for the best graduating student
- 04/2013–09/2014 **Junior studies in mathematics**, *Fernuniversität Hagen and PFH Göttingen*
06/2014 **Abitur**, *Joachim-Hahn-Gymnasium Blaubeuren*

Awards and Scholarships

- 06/2026 DFG Walter Benjamin Fellowship for a research project at the University of Cambridge, hosted by Prof. Clément Mouhot (ca. 100,000 EUR)
- 02/2025 PhD thesis award “Promotionspreis der Ulmer Universitätsgesellschaft” (1500 EUR)
- 06/2022 Travel grant from the ProTrainU Mobility-Programme, Ulm University (1000 EUR)
- 04/2022 Master’s graduation award “Absolventenpreis M.Sc. Mathematik Uni Ulm 2020” for the best graduation (500 EUR)
- 11/2019 Graduate scholarship granted by the State of Baden-Wuerttemberg, Germany (52,200 EUR)
- 05/2018 Bachelor’s graduation award “Absolventenpreis B.Sc. Mathematik Uni Ulm 2018” for the best graduation (500 EUR)
- 06/2014 Ferry Porsche award and graduation award of the Deutsche Physikalische Gesellschaft

Research

Preprints

- [1] **An autonomous Lipschitz fast dynamo on the three-torus.** arXiv:2608.02586, 2026.
- [2] **Sharp kinetic trace theory** with Lisa Valentini. arXiv:2607.24708, 2026.
- [3] **On the kinetic p -Laplace equation with nonlocal diffusion.** arXiv:2605.21080, 2026.
- [4] **Nonlinear kinetic diffusion equations with p -growth** with Helge Dietert and Rico Zacher. arXiv:2605.18521, 2026.
- [5] **Transfer of regularity by kinetic mollification along critical trajectories.** arXiv:2605.13582, 2026.
- [6] **Spherical rigidity for an exterior overdetermined problem with Neumann data prescribed by mean curvature.** arXiv:2604.07002, 2026.
- [7] **Kinetic Sobolev Spaces** with Pascal Auscher. arXiv:2603.17491, 2026.

- [8] **Nash's G bound for the Kolmogorov equation** with Helge Dietert. arXiv:2510.21621, 2025.
- [9] **Global rigidity of two-dimensional bubbles.** arXiv:2510.17557, 2025.
- [10] **Critical trajectories in kinetic geometry** with Helge Dietert, Clément Mouhot, and Rico Zacher. arXiv:2508.14868, 2025.
- [11] **Weak solutions to Kolmogorov–Fokker–Planck equations: regularity, existence and uniqueness** with Pascal Auscher and Cyril Imbert. arXiv:2403.17464, 2024.

Publications

- [12] **Steady bubbles and drops in inviscid fluids** with David Meyer and Christian Seis. *Calc. Var.*, 64(299), 2025.
- [13] **A trajectorial interpretation of Moser's proof of the Harnack inequality** with Rico Zacher. *Ann. Sc. Norm. Super. Pisa Cl. Sci. (5)*, 26(2):605–617, 2025.
- [14] **Fundamental solutions to Kolmogorov–Fokker–Planck equations with rough coefficients: existence, uniqueness, upper estimates** with Pascal Auscher and Cyril Imbert. *SIAM J. Math. Anal.*, 57(2):2114–2137, 2025.
- [15] **On a kinetic Poincaré inequality and beyond** with Rico Zacher. *J. Funct. Anal.*, 289(1):Paper No. 110899, 18, 2025.
- [16] **Analytic aspects of kinetic partial differential equations.** *OPARU — Open Access Repository der Universität Ulm*, 2023.
- [17] **Kinetic maximal L^p -regularity with temporal weights and application to quasilinear kinetic diffusion equations** with Rico Zacher. *J. Differential Equations*, 307:29–82, 2022.
- [18] **Kinetic maximal $L^p_\mu(L^p)$ -regularity for the fractional Kolmogorov equation with variable density.** *Nonlinear Anal.*, 214:Paper No. 112517, 21, 2022.
- [19] **Kinetic maximal L^2 -regularity for the (fractional) Kolmogorov equation** with Rico Zacher. *J. Evol. Equ.*, 21(3):3585–3612, 2021.

Employment History

- 10/2022–03/2023 **Predoctoral researcher**, *Ulm University*
- 03/2020–10/2022 **Teaching assistant**, *University of Applied Sciences Neu-Ulm*
- 10/2015–10/2019 **Teaching assistant**, *Ulm University*
- 03/2015–03/2019 **Research assistant (working student)**, *Daimler AG Research & Development Ulm*

Teaching Experience

University of Münster

- 2023–2025 Teaching assistant for *Partial Differential Equations* (1 sem.) and *Analysis and Numerics of Differential Equations* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam support)

- Winter 2024/25 Supervised a student seminar on *De Giorgi–Nash–Moser Theory*

- 2024–2025 Examined two BSc theses; co-supervised and examined the MSc thesis of Yuanjiang Han (now a PhD student with Christian Seis)

University of Applied Sciences Neu-Ulm

- 2020–2022 Teaching assistant for *Business Mathematics and Statistics I* (3 sem.) and *Business Mathematics and Statistics II – Regression Models / Statistical Testing* (2 sem.) (authored exercise sheets; ran the programming lab; marked submissions; created teaching videos)

Ulm University

- 2017–2022 Teaching assistant for *Analysis I* (2 sem.), *Analysis II* (1 sem.), *Dynamical Systems* (2 sem.), and *Elements of Functional Analysis* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam-setting support)

- Winter 2022/23 Local coordinator of the 26th Internetseminar

- Winter 2019/20 Organised the “Trainingscamp”, an annual four-week maths skills refresher for first-year students
2015–2017 Matlab tutor for *Analysis I* and *Analysis II*

Talks and Posters

- 05/2026 Talk – Calderón–Zygmund Analysis Seminar, University of Chicago
05/2026 Talk – Seminar Applied and Computational Mathematics, Tulane University
04/2026 Talk – ISU Analysis and Probability Seminar, Iowa State University
03/2026 Talk – Horizons in nonlinear PDEs, Ulm University
10/2025 Talk – Oberseminar Angewandte Analysis, Dortmund University
10/2025 Talk – Noko25 43rd Northern German Colloquium, Humboldt-Universität zu Berlin
09/2025 Poster – Hyperbolic Problems in Nürnberg 2025, FAU Erlangen
05/2025 Talk – Seminar Nichtlineare Partielle Differentialgleichungen, KIT Karlsruhe
03/2025 Poster – Kinetic equations and turbulence, CentraleSupélec and Collège de France
10/2024 Talk – Münster–Imperial Day in PDE, University of Münster
09/2024 Talk – Oberseminar Analysis, University of Bonn
03/2024 Poster – Mathematics Münster Mid-term Conference, University of Münster
11/2023 Talk – Séminaire Analyse Numérique et EDP, Université Paris-Saclay
09/2023 Talk – Seminar of the AG Analysis-Probability, MPI Leipzig
06/2023 Talk – Real Analysis and Geometry, CIRM Marseille
04/2023 Talk – Wilhelm Killing Colloquium, University of Münster
03/2023 Talk – Chair Dynamics Control and Numerics, FAU Erlangen
02/2023 Talk – TULKKA, KIT Karlsruhe
11/2022 Poster – Kinetic Theory, CIRM Marseille
09/2022 Talk & Poster – Horizons in non-linear PDEs, Ulm University
07/2022 Poster – When Kinetic Theory meets Fluid Mechanics, ETH Zürich
06/2022 Poster – 11th Summer school on “Methods and Models of Kinetic Theory”, Pesaro
01/2022 Poster – Frontiers in analysis of kinetic equations, INI Cambridge
07/2021 Talk – Nonlinear Evolution Equations and Approximations, University of Duisburg-Essen
03/2021 Talk – Oberseminar Analysis, Ulm University
02/2021 Poster – Winterschool on Analysis and Applied Mathematics, University of Münster

Research Stays

- 05/2026 Visit to Luis Silvestre (University of Chicago), one week
04/2026 Visit to Sam Punshon-Smith (Tulane University), one week
04/2026 Visit to David Herzog (Iowa State University), two weeks
2023–2025 Multiple visits to Pascal Auscher (Université Paris-Saclay), totalling 7 weeks
2023–2025 Multiple visits to Rico Zacher (Ulm University), totalling 5 weeks
2021–2025 Three visits to Cyril Imbert (École normale supérieure, Paris), totalling two weeks
05/2025 Visit to Björn de Rijk (KIT Karlsruhe), one week
09/2023 Visit to Tobias Ried (MPI Leipzig), one week
03/2023 Visit to Nicola De Nitti (FAU Erlangen), one week
02/2023 Visit to Christian Seis (University of Münster), half a week

Other Scientific Activities

- 2024 & 2026 Supporting the organisation of the Münster–Imperial Day in PDE, University of Münster
2022 Supporting the organisation of the summer school “Horizons in non-linear PDEs” at Ulm University
2020 Co-organisation of the seminar “Mathematisches Kolloquium” at Ulm University for six semesters
2022 Organisation of a weekend workshop for mathematics PhD students at Ulm University: 15 participants, secured 2500 EUR in funding from StuVe Ulm

- 2022 Supervision of a three-month project for the workshop of the 25th Internetseminar. Title: Kolmogorov equation in $L^p(\mathbb{R}^{2n})$ – Semigroup, spectrum, maximal regularity?
- 2021 Facilitator at the autumn school “COLLAB – Collaboratory for Global Sustainability 2021”, Toronto/Ulm
- 2019 Supporting the organisation of the winter school “Grad. Flows and Var. Methods in PDEs” at Ulm University
- Active contribution to the PDE seminars at the University of Münster and Ulm University by inviting several external speakers
- Attended over 30 international workshops and conferences, including events at CIRM Marseille, INI Cambridge, ETH Zürich, and the University of Oxford
- Peer reviewer for *Annales scientifiques de l'École normale supérieure*, *Communications in Partial Differential Equations*, *Discrete and Continuous Dynamical Systems*, *European Journal of Mathematics*, *Journal of Evolution Equations*, *Mathematika*, *Nonlinearity* and *Potential Analysis*

Academic Volunteer Work

- 02/2021–09/2022 **Elected member of the council of PhD students**, *Ulm University*
Representing the interests of doctoral students in university committees, e.g. for the change of the PhD regulations; chair of the council of PhD students (12/2021–09/2022)
- 02/2021–09/2022 **Elected member of the faculty council for Mathematics and Economic Sciences**, *Ulm University*
- 07/2017–03/2019 **Member of the Mathematics Student Council**, *Ulm University*
Member of the appointment committee for a professorship in “Stochastics”